

Exponential ExpressionsQuestion Bank 3.03 **ANSWER KEY***Show your work. Circle your final answer.*

1. Two students evaluated $\frac{(5^2 \cdot 2.8^0)^4}{5^7}$. Patricia first wrote 2.8^0 as 1, then $(5^2)^4 = 5^8$, then $\frac{5^8}{5^7} = 5$. Kim first wrote $(5^2)^4 \cdot (2.8^0)^4 = 5^8 \cdot 2.8^0$, then $5^8 \cdot 1$, then $\frac{5^8}{5^7} = 5$.

- a. Name the law Patricia used first, and the law she used second.

zero exponent, then power of a power

- b. Name the law Kim used first.

power of a product (the exponent 4 went to both factors)

- c. Name the law both students used last, and give the value of the expression.

quotient of powers; the value is 5

For questions 2 through 5, use the laws of exponents to evaluate. Show each law you use.

2. $\left(\frac{1.5^5}{1.5^3}\right)^2$
 $(1.5^2)^2 = 1.5^4 = 5.0625$

3. $\left[\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^1\right]^2$
 $\left(1 \cdot \frac{4}{5}\right)^2 = \frac{16}{25}$

4. $4.7^2 \cdot (4.7^1)^3$
 $4.7^2 \cdot 4.7^3 = 4.7^5 = 2,293.45007$

5. $\frac{9.9}{8.8^0} \cdot 9.9^3$
 $\frac{9.9}{1} \cdot 9.9^3 = 9.9^4 = 9,605.9601$

For questions 6 and 7, evaluate. Show each step.

6. $\frac{(2^2 \cdot 3.2^0)^6}{2^{10}}$
 $\frac{2^{12}}{2^{10}} = 2^2 = 4$

$$7. \quad \frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$$\frac{(-4)^3}{1} = -64$$

Equivalent or not.

8. Select ALL of the expressions that are equivalent to $\frac{0.2^3}{2^3} \cdot 0.1^2$.

- ☒ A. 0.1^5
- ☒ B. $0.1^3 \cdot 0.1^2$
- ☐ C. $\frac{0.2^5}{0.2^3}$
- ☒ D. $\left(\frac{0.2}{2}\right)^3 \cdot 0.1^2$
- ☐ E. $0.2^3 \cdot 0.1^2$
- ☐ F. $0.2^0 \cdot 0.1^2$

A, B and D

The expression is $0.1^3 \cdot 0.1^2 = 0.1^5 = 0.00001$.

9. Which expression is equivalent to $\left(\frac{3^5}{3^2}\right)^2$?

- ☐ A. 3^5
- ☒ B. 3^6
- ☐ C. 3^{14}
- ☐ D. 3^7

B

3^3 squared is $3^6 = 729$.

Working backwards.

10. Find the value of n that makes each statement true.

a. $\left(\frac{2^7}{2^n}\right)^2 = 2^6$

$n = 4$

Inside must be 2^3 .

b. $\frac{5^n \cdot 5^0}{5^3} = 25$

$n = 5$

$5^{n-3} = 5^2$.

11. Ruth says that $\frac{(-15)^4}{1.5^4}$ can be rewritten as $\left(\frac{-15}{1.5}\right)^4$ before evaluating. Is she right? Name the law, and give the value.

Yes — power of a quotient, read from right to left; $(-10)^4 = 10,000$.